Nested-logit demand: troubleshooting sigma above one

October 14, 2026

1. Problem and decisions needed

The IV nesting parameter exceeds one in the baseline demand equation:

$$\log(s_{it}/s_{0t}) = c + \alpha p_{it} + \beta_a a_{it} + \sigma \log(s_{it}/s_{gt}) + \xi_{it}.$$

Here i indexes incumbent brands and t indexes months. s_{it} is the brand's share of potential market volume, s_{0t} is the outside share, and s_{gt} is total e-cigarette share, including incumbents and fringe. p_{it} is real price per mL, a_{it} is delivered nicotine concentration, and ξ_{it} captures unobserved demand attributes. The signed price coefficient α is expected to be negative; the maintained nesting range is $0 \leq \sigma < 1$.

Retail corrections, firm fixed effects, additional product controls, and alternative market-size constructions have not resolved the out-of-range estimate.

Potential threats: (i) outside option; (ii) exclusion violation; (iii) conditioning on nicotine or capacity reverses the national tax–price association; and (iv) aggregation, common changes over time, or endogenous characteristics may explain that sensitivity.

2. Market, observations, and variables

The demand market is national e-cigarette retail sales in a calendar month, January 2013–December 2023. All individually modeled incumbent brands and the aggregate fringe are in one e-cigarette nest; the outside alternative is a singleton.

Eligible UPC-months have a known brand, positive units and liquid volume, and at least three reporting stores nationally in the month. Eligible non-incumbent brands are pooled into fringe volume. Fringe sales enter total e-cigarette volume and the share denominators, but the fringe is not a separate i to be estimated in the demand regression.

An incumbent is a brand with **annual eligible liquid-volume share above 7%**, with its assignment fixed within the year. This demand roster is distinct from the dynamic model's proposed cap of five firms: that will come into play in the supply side. The full estimation sample contains 540 incumbent brand-months. For comparisons involving capacity, BLU's capacity measure is unresolved in all 12 months of 2018. These observations are omitted from the matched estimation sample, leaving 528 observations. All matched specifications use these same 528 rows, including specifications without capacity. BLU remains an incumbent, and its eligible sales remain in total e-cigarette volume when constructing every other brand's shares.

Price and nicotine concentration are averaged over all eligible UPCs, including bottles and mixed product types. Capacity uses only eligible cartridge/pod and disposable UPCs with a defined per-device capacity. Thus the capacity average can cover a narrower assortment than price and nicotine.

Variable	Construction
q_{it}	Incumbent brand's eligible liquid volume, mL
Q_t	Total eligible liquid mL across incumbents and fringe
p_{it}	unweighted UPC mean of UPC-month real revenue divided by corrected mL
a_{it}	unweighted-UPC delivered nicotine, mg/mL, using the existing 0.68 transfer factor
Capacity	unweighted-UPC mean mL per cartridge/disposable; excludes bottles and mixed types
Variety	Log number of distinct eligible UPCs, including bottles

Representative price times firm volume generally does not equal observed revenue. This matters for later profit measurement.

3. Estimating equation and maintained assumptions

$$\log(s_{it}/s_{0t}) = c + \alpha p_{it} + \beta_a a_{it} + \sigma \log(s_{it}/s_{gt}) + \gamma_C C_{it} + \gamma_V V_{it} + \mu_i + \xi_{it}.$$

Capacity, variety and firm effects appear only where specified in alternative attempts. Nicotine is included in every structural demand specification reported below. Nicotine, capacity and variety are treated as exogenous; **this remains an assumption.**

IV instruments endogenous **price and within-nest log share** with the two excluded instruments defined below. Both instruments enter both first stages.

All structural fits use month-clustered standard errors (132 clusters). I don't include month FE, because with one inside nest,

$$\log(s_{it}/s_{0t}) = \log(s_{it}/s_{gt}) + \log(s_{gt}/s_{0t}).$$

Full month effects absorb the second term and create a degeneracy for identifying the intended demand relationship. Month clustering does not address serial dependence across months.

4. Instrument definitions

Instrument 1 — State-population-weighted tax

Let S_{it} be the set of states where brand i has positive sales of a nationally eligible UPC in month t. The instrument is

$$T_{it} = \sum_{s \in S_{it}} \frac{P_{s,y(t)}}{\sum_{r \in S_{it}} P_{r,y(t)}} \tau_{st}.$$

Here P is annual state population and τ_{st} is the saved closed-system excise-tax rate in real dollars per mL in state s. The weights are population weights. Store eligibility is imposed nationally; three stores in each state are not required. Both tax changes and changes in a firm's sales footprint can change this instrument. This **may be endogenous**, and the closed-system tax may imperfectly represent open-system/refill products.

Instrument 2, alternative A — National active-brand count

$N_t^{national}$ is the number of **distinct eligible active brands anywhere in the US in month** t. A brand qualifies through its eligible positive-sales UPCs; a brand with many UPCs counts once. This count includes the focal brand, other incumbents, and fringe brands (for ex can include 10 fringe brands). It is not the incumbent count, UPC count, number of states, or number of parent companies. It is common across focal firms within a month.

Instrument 2, alternative B — Active-brand count across the firm's operating states

N_{it}^{geo} counts distinct eligible brands with positive sales in at least one state in S_{it}, the union of states where focal brand i operates. Each competitor counts once across that union. This also includes the focal brand and fringe brands. It does not require that a competitor operate in every focal-firm state.

$N_t^{national} = N_{it}^{geo}$ **in 528 of the 540 firm-months**, meaning that in those observations, every nationally eligible brand sells in at least one state where the focal firm i sells.

The baseline specifications use Instrument 1 together with Instrument 2A. Alternative specifications replace 2A with 2B while retaining the same tax instrument. In the tables below, "national" and "geographic" refer to these two brand-count alternatives.

5. Market size and outside option

Let A_y be annual US adult population (July estimate held fixed within year). NCP supplies monthly purchasing intensity

$$k_t = \frac{\sum_{h \in H_t} w_{ht} v_{ht}}{\sum_{h \in H_t} w_{ht}},$$

where H_t contains e-cigarette-purchasing households with usable volume, v_{ht} is monthly purchased mL, and w_{ht} is the projection weight. Thus k_t is **conditional on household purchase**, not an average over all households or all adults.

Attempted definitions	Potential volume M_t
Annual baseline	Adult population times arithmetic mean of monthly k_t within year
Monthly intensity	Adult population times current month's k_t
Constant intensity	Adult population times arithmetic mean of k_t over all 132 months

Constant intensity still permits annual changes in population.

$$s_{it} = q_{it}/M_t, \qquad s_{gt} = Q_t/M_t, \qquad s_{0t} = 1 - Q_t/M_t.$$

$$\log(s_{it}/s_{0t}) = \log\left(\frac{q_{it}}{M_t - Q_t}\right), \qquad s_{it}/s_{gt} = q_{it}/Q_t.$$

Changing M_t changes the demand outcome and outside share, but not within-nest share. All outside shares are positive, approximately 98.91%–99.93% across alternatives.

Interpretation unresolved: the denominator multiplies purchasing-household intensity by adults, while the numerator is scanner-covered eligible retail volume. Household/person units and channel coverage may need alignment. The outside share is residual potential volume, not the fraction of adults who do not vape. Changing the denominator alone cannot convert mL shares to adult participation shares. No alternative denominator is approved merely because it yields a smaller sigma.

6. Complete structural specification inventory

I tried 60 distinct structural specifications: 12 full-sample models plus 48 matched-sample (due to BLU issue) market-size models.

12 Full-sample baseline: M1–M6 and D1–D6

All use 540 observations, annual baseline market size, nicotine, no capacity or variety. M1–M4 are full IV. M5/M6 are OLS. D1–D4 instrument share only and treat price as exogenous; D5/D6 instrument price only and treat share as exogenous. The untreated endogenous candidate enters the other regressor's first stage as an ordinary control.

Model	Estimator	Firm FE	Excluded instruments	Alpha	Sigma
M1	Full IV	No	Tax + national count	-0.211	3.422
M2	Full IV	No	Tax + geographic count	-0.202	3.630
M3	Full IV	Yes	Tax + national count	-0.339	4.024
M4	Full IV	Yes	Tax + geographic count	-0.332	4.159
M5	OLS	No	None	-0.035	0.975
M6	OLS	Yes	None	-0.046	0.911
D1	Share-only IV	No	National count	-0.461	5.699

Model	Estimator	Firm FE	Excluded instruments	Alpha	Sigma
D2	Share-only IV	No	Geographic count	-1.349	15.538
D3	Share-only IV	Yes	National count	-0.263	3.536
D4	Share-only IV	Yes	Geographic count	-0.292	3.883
D5	Price-only IV	No	Tax	-0.303	1.345
D6	Price-only IV	Yes	Tax	-0.490	1.130

Some partial-IV first stages are very weak: share-only IV excluded-count F statistics range from 0.04 to 8.06. Without firm fixed effects, the share-only IV first-stage F statistics are particularly low: 0.46 with national count (D1) and 0.04 with geographic count (D2); adding firm fixed effects raises them to 8.06 (D3) and 5.45 (D4), respectively. This is a different diagnostic from conditional strength in full IV.

Matched full-IV specifications: B, S, V, SV

All use 528 observations and include nicotine. B adds neither control; S adds capacity; V adds log UPC variety; SV adds both. Each family has four instrument/FE settings:

Suffix	Count instrument	Firm FE
1	National	No
2	Geographic	No
3	National	Yes
4	Geographic	Yes

Every one of B1–B4, S1–S4, V1–V4 and SV1–SV4 is estimated under annual, monthly and constant intensity market definitions: $16 \times 3 = 48$ specifications. The tables below enumerate all 48 and show sigma and signed alpha. ### Sigma: all market-size variants

Model	Controls	FE	Count	Annual	Monthly	Constant
B1	Baseline	No	National	2.887	2.773	3.103
B2	Baseline	No	Geographic	3.036	2.902	3.296
B3	Baseline	Yes	National	2.384	2.255	2.952
B4	Baseline	Yes	Geographic	2.409	2.271	2.999
S1	Size	No	National	2.880	2.771	2.916
S2	Size	No	Geographic	3.045	2.914	3.144
S3	Size	Yes	National	2.267	2.160	2.641
S4	Size	Yes	Geographic	2.299	2.181	2.705
V1	Variety	No	National	3.316	3.142	3.979
V2	Variety	No	Geographic	3.487	3.291	4.192
V3	Variety	Yes	National	2.315	2.169	3.054
V4	Variety	Yes	Geographic	2.347	2.191	3.099
SV1	Size and variety	No	National	3.281	3.123	3.469
SV2	Size and variety	No	Geographic	3.491	3.304	3.759
SV3	Size and variety	Yes	National	2.186	2.067	2.683
SV4	Size and variety	Yes	Geographic	2.227	2.096	2.748

Signed alpha: all market-size variants

Model	Controls	FE	Count	Annual	Monthly	Constant
B1	Baseline	No	National	-0.113	-0.105	-0.302
B2	Baseline	No	Geographic	-0.103	-0.097	-0.289

Model	Controls	FE	Count	Annual	Monthly	Constant
B3	Baseline	Yes	National	-0.002	0.008	-0.294
B4	Baseline	Yes	Geographic	0.005	0.012	-0.281
S1	Size	No	National	-0.115	-0.106	-0.359
S2	Size	No	Geographic	-0.100	-0.093	-0.338
S3	Size	Yes	National	-0.024	-0.010	-0.351
S4	Size	Yes	Geographic	-0.015	-0.004	-0.334
V1	Variety	No	National	-0.189	-0.171	-0.458
V2	Variety	No	Geographic	-0.187	-0.169	-0.455
V3	Variety	Yes	National	0.011	0.025	-0.314
V4	Variety	Yes	Geographic	0.019	0.030	-0.303
SV1	Size and variety	No	National	-0.189	-0.171	-0.462
SV2	Size and variety	No	Geographic	-0.187	-0.169	-0.457
SV3	Size and variety	Yes	National	-0.005	0.012	-0.361
SV4	Size and variety	Yes	Geographic	0.004	0.019	-0.346

7. Instrument-strength assessment

Sanderson–Windmeijer conditional F diagnostics were computed for price and within-nest share in M1–M4 and all 16 matched settings, accounting for the other endogenous regressor. Their first-stage data do not change with market size, so the same matched diagnostics apply to all 48 fits.

Full-IV model	Price conditional F	Share conditional F
M1	55.79	62.68
M2	58.65	49.08
M3	15.91	18.95
M4	15.08	17.91
B3	75.88	92.09
B4	77.36	91.82
SV3	81.38	79.57
SV4	82.31	78.38

Across all matched settings, the month-clustered Sanderson–Windmeijer conditional first-stage F statistics range from 40.39 to 164.50. There is no obvious collapse of conditional relevance in the matched full-IV fits. **The change from M3/M4 to B3/B4 highlights influence of the excluded BLU 2018 observations even before adding controls.** Serial dependence and weak-IV-robust inference remain open.

8. Odd tax–price sign and conditioning on characteristics

On the same 528 observations, with geographic tax, national count and firm FE, the price-first-stage tax slope changes as follows. **These are price-regression diagnostics**, not additional structural demand fits.

Included characteristics	Tax slope	Month-clustered SE
None	+0.84379	0.29271
Variety only	+0.77509	0.28492
Nicotine only	-1.43218	0.33995
Capacity only	-1.10000	0.28167
Nicotine + capacity	-1.67242	0.34956
Nicotine + variety	-1.41189	0.33129
Capacity + variety	-1.08758	0.28047
Nicotine + capacity + variety (SV3)	-1.69852	0.33985

Removing nicotine and capacity **together** reverses the negative sign. Adding either **separately** to the otherwise basic first stage produces a negative slope. Removing only one from SV3 leaves it negative.

Nicotine and capacity may reflect firms' responses to unobserved demand, but correlated exogenous attributes, common changes over time and product composition can also produce sign reversals. Removing a core demand attribute solely to make tax positive would not establish a valid specification. Treating these attributes as endogenous would require additional excluded instruments and joint identification.

For SV3, tax coefficients in the price first stage, share first stage and outcome reduced form are -1.69852, +0.75910 and +1.66828. α is approximately -0.00503 and σ 2.18644.

9. Aggregation diagnostics and checks already completed

The original state-level diagnostic uses all eligible brands: 6,443 state-months and 699,282 UPC-state-months. Tax slopes are positive for equal-UPC state prices with no FE, state FE, month FE, or both. With both state and month FE, the slope is +0.713; with UPC-state pair and month FE, +0.835.

The tighter matched bridge retains exactly the 528 SV3/SV4 incumbent firm-months and expands them to 25,044 operating-state observations. State regressions retain national firm-month nicotine/capacity/variety controls. Default observation weights give each firm-month total weight one across states.

Matched local price regression	Tax slope	SE
Firm FE + count + characteristics	+0.923	0.072
Firm + state FE + count + characteristics	+0.641	0.157
Firm + state + month FE + characteristics	+0.841	0.149
Unweighted rows, firm + state FE + controls	+0.634	0.158
Volume-price sensitivity, firm + state + month FE	+0.887	0.107

State errors are clustered by state and month. National count is absorbed by month FE in the diagnostic. These price regressions can include month FE without the structural single-nest demand degeneracy.

National sensitivity fits cross five price constructions (current pooled-UPC price; equal-, population-, or sales-weighted local equal-UPC prices; aggregate revenue/volume) with three tax weights (population, equal-state, current mL sales). All 15 combinations retain 528 rows, national count, the three characteristics and firm FE. All tax slopes are negative.

10. Questions for discussion

1. Should the economic choice unit be adult participation or volume/purchase opportunities? What denominator and scanner-coverage adjustment are defensible?
2. What supports exclusion for geographic tax and active-brand count given endogenous entry, footprints and product choice?
3. Does nicotine/capacity conditioning remove confounding or create identification problems through endogenous characteristics? What additional instruments could credibly identify them if so?
4. Should demand be modeled geographically or at a less aggregated product level? What is lost by the representative firm price and fringe aggregation?
5. Which next diagnostic is most informative: leave-firm/year-out influence, time-pattern analysis, serial-dependence-robust strength, or weak-IV-robust joint inference for price and sigma?

Appendix A. Full coefficients and standard errors for annual models

Entries below reproduce the saved comparison tables. Alpha is signed; parentheses are month-clustered standard errors. Ordinary first-stage F values shown here are joint excluded-instrument tests, not the conditional F statistics in Section 7.

Full-sample main models

	M1	M2	M3	M4	M5	M6
Estimator	Full IV	Full IV	Full IV	Full IV	OLS	OLS
Firm FE	No	No	Yes	Yes	No	Yes
Count IV	National	Geographic	National	Geographic	–	–
Tax IV	Yes	Yes	Yes	Yes	No	No
alpha: price	-0.211 (0.117)	-0.202 (0.122)	-0.339 (0.220)	-0.332 (0.225)	-0.035 (0.007)	-0.046 (0.016)
beta_a: nicotine	0.007 (0.013)	-0.000 (0.016)	0.037 (0.010)	0.035 (0.011)	0.055 (0.003)	0.057 (0.005)
sigma: nesting	3.422 (0.665)	3.630 (0.769)	4.024 (1.087)	4.159 (1.162)	0.975 (0.057)	0.911 (0.052)
N	540	540	540	540	540	540
Price first-stage F	62.93	63.11	36.32	36.81	–	–
Share first-stage F	55.00	39.17	48.63	37.91	–	–
sigma in [0,1)	No	No	No	No	Yes	Yes

Partial IV

	D1	D2	D3	D4	D5	D6
Estimator	Share-only IV	Share-only IV	Share-only IV	Share-only IV	Price-only IV	Price-only IV
Firm FE	No	No	Yes	Yes	No	Yes
Count IV	National	Geographic	National	Geographic	–	–
Tax IV	No	No	No	No	Yes	Yes
alpha: price	-0.461 (0.642)	-1.349 (6.524)	-0.263 (0.103)	-0.292 (0.131)	-0.303 (0.056)	-0.490 (0.109)
beta_a: nicotine	-0.025 (0.117)	-0.193 (1.216)	0.038 (0.010)	0.035 (0.011)	0.082 (0.008)	0.089 (0.012)
sigma: nesting	5.699 (7.026)	15.538 (71.921)	3.536 (1.165)	3.883 (1.520)	1.345 (0.058)	1.130 (0.062)

	D1	D2	D3	D4	D5	D6
N	540	540	540	540	540	540
Price first-stage F	–	–	–	–	62.57	37.91
Share first-stage F	0.46	0.04	8.06	5.45	–	–
sigma in [0,1)	No	No	No	No	No	No

Matched baseline

	B1	B2	B3	B4
Firm FE	No	No	Yes	Yes
Count IV	National	Geographic	National	Geographic
alpha	-0.113 (0.065)	-0.103 (0.069)	-0.002 (0.070)	0.005 (0.071)
beta_a	0.013 (0.010)	0.007 (0.011)	0.036 (0.007)	0.035 (0.007)
beta_size	–	–	–	–
beta_variety	–	–	–	–
sigma	2.887 (0.356)	3.036 (0.408)	2.384 (0.231)	2.409 (0.234)
N	528	528	528	528
p_it first-stage F	74.35	74.60	35.30	35.78
ln_s_i_g first-stage F	73.52	51.21	84.50	70.46

Size

	S1	S2	S3	S4
Firm FE	No	No	Yes	Yes
Count IV	National	Geographic	National	Geographic
alpha	-0.115 (0.081)	-0.100 (0.085)	-0.024 (0.075)	-0.015 (0.076)
beta_a	0.012 (0.009)	0.008 (0.010)	0.015 (0.009)	0.015 (0.009)
beta_size	-0.018 (0.146)	0.024 (0.155)	-0.214 (0.091)	-0.208 (0.091)
beta_variety	–	–	–	–
sigma	2.880 (0.340)	3.045 (0.400)	2.267 (0.190)	2.299 (0.195)
N	528	528	528	528
p_it first-stage F	68.21	68.72	34.95	35.28
ln_s_i_g first-stage F	53.86	36.58	70.61	67.22

Variety

	V1	V2	V3	V4
Firm FE	No	No	Yes	Yes
Count IV	National	Geographic	National	Geographic
alpha	-0.189 (0.095)	-0.187 (0.100)	0.011 (0.065)	0.019 (0.066)
beta_a	0.006 (0.014)	0.001 (0.015)	0.040 (0.007)	0.039 (0.007)
beta_size	–	–	–	–
beta_variety	-0.483 (0.182)	-0.523 (0.202)	0.134 (0.103)	0.130 (0.106)

	V1	V2	V3	V4
sigma	3.316 (0.563)	3.487 (0.641)	2.315 (0.236)	2.347 (0.241)
N	528	528	528	528
p_it first-stage F	84.32	82.61	36.67	36.94
ln_s_i_g first-stage F	52.10	40.79	67.65	57.27

Size and variety

	SV1	SV2	SV3	SV4
Firm FE	No	No	Yes	Yes
Count IV	National	Geographic	National	Geographic
alpha	-0.189 (0.095)	-0.187 (0.101)	-0.005 (0.067)	0.004 (0.068)
beta_a	0.006 (0.013)	0.001 (0.015)	0.022 (0.009)	0.021 (0.009)
beta_size	-0.033 (0.166)	0.003 (0.175)	-0.201 (0.084)	-0.194 (0.085)
beta_variety	-0.459 (0.185)	-0.525 (0.213)	0.173 (0.099)	0.167 (0.102)
sigma	3.281 (0.528)	3.491 (0.623)	2.186 (0.192)	2.227 (0.198)
N	528	528	528	528
p_it first-stage F	69.77	69.73	36.50	36.83
ln_s_i_g first-stage F	34.15	27.61	60.12	55.49

Appendix B. Diagnostic specification inventory and saved results

The following tables list every saved matched bridge price regression (including duplicate benchmark constructions for reconciliation) and the broad state-level diagnostic. They are auxiliary OLS fits, not additional nested-logit estimates. N is the observation count.

Matched bridge

Diagnostic	N	Tax slope	SE
N current SV3	528	-1.6985	0.3399
N current SV4	528	-1.6954	0.3420
N current without product controls	528	0.8438	0.2927
N controls a it	528	-1.4322	0.3399
N controls avg ml per ecig	528	-1.1000	0.2817
N controls a it avg ml per ecig	528	-1.6724	0.3496
N controls log upc variety	528	0.7751	0.2849
N controls a it log upc variety	528	-1.4119	0.3313
N controls avg ml per ecig log upc variety	528	-1.0876	0.2805
N p it tax pop	528	-1.6985	0.3399
N p it tax equal	528	-1.3947	0.4508
N p it tax sales	528	-1.6662	0.7891
N p state equal tax pop	528	-0.7594	0.4631
N p state equal tax equal	528	-2.5660	0.5468
N p state equal tax sales	528	-1.8402	1.0354
N p state pop tax pop	528	-0.7580	0.4719
N p state pop tax equal	528	-2.5465	0.5344
N p state pop tax sales	528	-1.8717	1.0594
N p state sales tax pop	528	-0.9193	0.4199
N p state sales tax equal	528	-2.4169	0.5216
N p state sales tax sales	528	-2.1207	1.0058
N p volume tax pop	528	-2.2452	0.4344

Diagnostic	N	Tax slope	SE
N p volume tax equal	528	-4.0770	0.2853
N p volume tax sales	528	-2.7712	0.6865
S brand count only	25044	0.9176	0.0716
S brand controls	25044	0.9227	0.0717
S brand STATE count only	25044	0.6354	0.1596
S brand STATE controls	25044	0.6406	0.1567
S brand STATE year month count only	25044	0.8394	0.1503
S brand STATE year month controls	25044	0.8409	0.1485
S state FE unweighted	25044	0.6336	0.1580
S state month FE volume price	25044	0.8871	0.1067

Broad state diagnostic

Diagnostic	N	Tax slope	SE
price pooled	6443	0.9365	0.0903
price state FE	6443	0.7789	0.1835
price month FE	6443	0.9242	0.0775
price state month FE	6443	0.7130	0.1619
price volume pooled	6443	0.7301	0.1179
price volume state FE	6443	0.5974	0.2089
price volume month FE	6443	0.8714	0.1460
price volume state month FE	6443	0.9859	0.1897
upc state month FE	699282	0.9568	0.1018
upc state pair month FE	699282	0.8348	0.1632

Additional completed checks

- All 40 conditional-F tests (20 settings × two regressors), including independent covariance/residual checks; market-size first-stage invariance verified.
- SV3/SV4: two first stages and one outcome reduced form each, with exact-identification coefficient decomposition.
- Retail rebuild audit, pack-volume identities, aggregation checks, independent QR OLS/2SLS coefficient and residual checks.
- An auxiliary policy-overlap exercise used 192 observed e-cigarette state-months in eight states. It crossed two price constructions, four FE settings (none/state/month/both), and policy inclusion/exclusion: 16 OLS fits. Limited coverage and absorbed policies prevent generalization to the full panel; this exercise did not change structural controls.

Reproducibility locations

Paths below are relative to `smoke_firm_dir/`.

- Structural estimates and standard errors: `output/demand_estimation/specification_comparison/README.md` and its `product_size/` and `market_size/` subdirectories.
- Conditional instrument strength: the comparison directory's `conditional_F/`.
- First stages and reduced forms: the comparison directory's `sv_first_stages_reduced_form/`.
- State-level diagnostics and matched bridge: `output/prelim_analysis/state_price_tax_diagnostic/`, including `aggregation_bridge/` and `policy_overlap/`.
- Diagnostic scripts: `code/2_preliminary_analysis/16_state_price_tax_diagnostic.R`, `17_state_policy_price_tax_` and `18_aggregation_price_tax_bridge.R`.

This meeting note summarizes existing saved estimates; no models were re-estimated to prepare it.

Price construction and retail-volume rebuild: consolidated reference

Consolidated from the standalone October 7 price-construction note. The October 20 follow-up below is later material, not part of the original October 14 presentation. Construction statements describe the status at their respective dates.

Original construction and October 7 rebuild

Firm-month price remains the UPC-unweighted mean of UPC-month prices in real dollars per mL:

`p_upc,t = sum(real revenue_upc,t) / sum(corrected mL sold_upc,t)`

`p_i,t = mean(p_upc,t across eligible UPCs belonging to firm i in month t)`

Collapse T2 rows to one brand × UPC × month before taking the ratio. Eligibility: UNKNOWN excluded, positive units and mL, and at least three reporting stores per UPC-month. Each eligible UPC has equal weight. The descriptive fringe price remains aggregate fringe revenue / mL.

Reviewed pack counts and capacities from input/manual_product_size_corrections.csv now feed the retail builders. The correction includes JUUL 081991301543 at one 0.7 mL pod, both reviewed MISTIC disposables at 1.5 mL, and Vuse Vibe at 1.9 mL per cartridge. The original JUUL UPC is retained under the user's approved information transfer. Nicotine quantities are recomputed using corrected liquid volume and existing clean concentration; the transfer factor remains 0.68.

Retail quantities, prices per mL, nicotine quantities, shares, annual incumbent eligibility, geographic instruments and downstream analyses are rebuilt. Raw revenue and unit observations are not edited. The separate NCP household purchasing-intensity input is unchanged by explicit user instruction.

Market size uses annual adult population (the July estimate held constant within each year) times either the annual mean, monthly value, or full-period mean of the saved household-weighted NCP monthly mL series. The annual-mean version remains the baseline.

The unweighted price times firm volume generally does not equal observed revenue: p_bar Q - R = sum_u (p_bar - p_u) q_u. Revenue per mL, R/Q, recovers revenue exactly. This remains a consideration for future profit measurement.

Reusable controls: average mL per cartridge/disposable (unweighted across eligible device UPCs) and log eligible UPC count. Refill bottles are excluded from device capacity but included in variety. Missing device size causes the full firm-month to be excluded from matched control comparisons; the unresolved BLU UPC is not imputed automatically. Current sample sizes and estimates are in output/demand_estimation/specification_comparison/README.md.

Implementation: code/1_build_data/8_build_firm_month_demand_panel.R and 10_build_product_size.R. Rebuild details and pre-rebuild backups are documented in the October 7 rebuild record.

Outside-option interpretation

The current volume-based outside share is not a non-vaping adult rate. Household purchasing intensity times adult population and scanner sales coverage require further assessment; see market-size definitions and the current handoff. No alternative denominator beyond the three existing variants has been adopted.

Later advisor follow-up: October 20, 2026

The separate sales-weighted price uses **e-liquid mL weights**: `p_it_volume_weighted = sum_u(Q_u * p_u) / sum_u Q_u = firm real revenue / firm mL`. Existing `p_it` remains UPC-unweighted. Both measures are saved for all eligible brand-months in `input/price_measures_firm_month.rds`; the demand builder includes the additional column on future runs. Existing canonical panels and demand fits were not overwritten.

Across 540 incumbent firm-months, existing `p_it * mL` totals $7.16248 billion versus $4.75711 billion in observed real revenue: net overstatement $2.40537 billion (50.56%). The median signed firm-month gap is +35.17%; the middle half spans +2.33% to +61.25%; the full range is -42.98% to +186.06%. Summed absolute dollar gaps equal 52.53%

of observed revenue. 419 observations overstate and 121 understate revenue. On the 528-row matched sample, net overstatement is 51.23%. These totals span 2013-2023 and refer to eligible scanner sales, not total industry revenue.

Full comparison contains firm breakdowns, with annual detail and largest discrepancies in accompanying CSVs. Volume-weighted price times volume reproduces observed revenue for every eligible brand-month. Reconstructed original price, volume and revenue match the saved 540-row incumbent panel. Reproduce with `Rscript --vanilla smoke_firm_dir/code/2_preliminary_analysis/16_price_revenue_comparison.R`. This comparison does not select a replacement demand specification.